

## Full length article

# Lightweight segmentation model for automated facade installation in high-rise buildings

Ahmed Yimam Hassen <sup>a</sup>, Mehrdad Arashpour <sup>b</sup>, Elahe Abdi <sup>a</sup>

<sup>a</sup> Department of Mechanical and Aerospace Engineering, Monash University, Clayton, VIC 3800, Australia

<sup>b</sup> Department of Civil Engineering, Monash University, Clayton, VIC 3800, Australia

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## ABSTRACT

The installation of curtain wall modules (CWM) in high-rise buildings is a complex task that poses significant safety risks due to manual labor, especially when working at great heights. Traditional methods are labor-intensive, time-consuming, and expose workers to hazards such as falls and equipment malfunctions. To mitigate these risks and enhance operational efficiency, automation and precise positioning of CWMs are essential. Accurate detection of installation locations becomes critical, as it enables crane operators or autonomous robots to position CWMs safely and precisely. This study introduces a novel approach utilizing semantic segmentation for detecting CWM installation locations. To address the challenges of deploying deep learning models on edge devices in construction environments, we propose Lightweight Attention Network (LANet), a lightweight, single-stream semantic segmentation architecture. LANet incorporates an optimized transformer module for global context modeling with linear complexity, enabling efficient feature extraction while maintaining computational efficiency. Additionally, we have developed a custom curtain wall dataset tailored for automating CWM installation, which was used to train and evaluate LANet. Experimental results demonstrate that LANet achieves competitive segmentation accuracy with only 1.92 million parameters, delivering real-time performance at 262 FPS on an RTX 3090 GPU and 19 FPS on a standard Intel i7 CPU. These results make LANet highly suitable for deployment in resource-constrained environments.

## 1. Introduction

In recent years, the rapid growth of high-rise construction [1,2] has introduced significant logistical and safety challenges in building processes [3,4]. This has driven demand for curtain walls, which provide a cost-effective, energy-efficient, and aesthetically versatile solution. A curtain wall is a prefabricated exterior façade, typically made of glass and other materials, that wraps wholly or partially around a metallic grid building structure. Curtain walls play a critical role in modern architecture and urban development, and their significance is expected to grow in the coming years. A recent market report projects that the global glass curtain wall market will expand from \$60.97 billion in 2024 to \$92.94 billion in 2029, at a compound annual growth rate of 8.9% [5]. Additionally, curtain wall construction accounts for 10%–15% of total building costs [6]. This growth is driven by several factors, including the role of curtain walls in urban heat island mitigation, their integration with building information modeling (BIM), and their compatibility with smart building technologies. Furthermore, curtain walls align with emerging trends in resilient building design and

net-zero energy buildings, reinforcing their importance in sustainable architecture.

Curtain walls are broadly classified into two types: the stick curtain wall and the unitized curtain wall. The stick curtain wall is assembled on-site, where prefabricated components – such as vertical and horizontal frames and glass or opaque spandrel panels – are installed piece by piece. This method offers flexibility but requires significant labor. In contrast, the unitized curtain wall consists of large, pre-assembled units of glass panels and spandrels that are fabricated off-site and then suspended from the main building structure. The unitized curtain wall relies on curtain wall modules (CWM)—pre-assembled façade units that integrate glass panels, spandrels, and mullions into a single structure. CWMs are manufactured under controlled conditions and transported to the construction site for direct installation. By consolidating multiple façade components into a single module, CWMs enhance installation efficiency, reduce on-site labor, and improve quality control compared to traditional stick-built systems.

\* Corresponding author.

E-mail addresses: [ahmed.hassen@monash.edu](mailto:ahmed.hassen@monash.edu) (A.Y. Hassen), [mehrdad.arashpour@monash.edu](mailto:mehrdad.arashpour@monash.edu) (M. Arashpour), [elahe.abdi@monash.edu](mailto:elahe.abdi@monash.edu) (E. Abdi).

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Despite their advantages, curtain wall module installation presents considerable challenges, particularly in high-rise construction. The process involves handling large and heavy components at high elevations, requiring precise alignment and coordination among workers operating heavy machinery. The conventional manual installation method is labor-intensive, time-consuming, and hazardous, exposing workers to risks such as falls, equipment malfunctions, and handling accidents. These challenges highlight the inefficiency of conventional methods and underscore the need for advanced automation to enhance safety and efficiency.

To address these challenges, researchers have explored automation solutions for façade construction and maintenance [3]. For instance, robotic systems have been proposed for high-risk tasks such as climbing robots for façade cleaning [7] and inspection [8], as well as unmanned aerial vehicles (UAVs) for façade assessment [9]. Additionally, indoor CWM installation robots [10] have been introduced to improve precision in panel alignment and placement, reducing reliance on manual labor. However, installing CWMs in high-rise buildings remains a major challenge, particularly in ensuring safety, efficiency, and precision [11]. A cable-driven parallel robot (CDPR) was developed for the installation of CWM in high-rise buildings, demonstrating competitive performance in terms of accuracy and installation time compared to conventional manual methods [12].

As discussed above several research efforts have focused on developing robotic systems for CWM installation [12,13]. However, relatively few studies have explored advanced computer vision solutions based on deep learning to enable robots to accurately identify installation positions and orientations [11,14]. Such perception capabilities are critical for autonomous CWM installation, as the required dimensional tolerances are typically in the millimeter range [12]. Deep learning has revolutionized machine perception in construction robotics [15–17], significantly advancing object detection for construction applications [18] and improving decision-making in autonomous systems [19, 20]. Developing robust deep learning-based semantic segmentation algorithm is essential to achieving precise detection and localization capabilities for autonomous CWM installation robots.

Early research in semantic segmentation primarily focused on CNN-based methods, which utilized encoding-decoding frameworks with convolution operations. Notable works in this domain include Fully Convolutional Networks (FCNs) [21], SegNet [22], UNet [23], PSPNet [24], OCRNet [25], the DeepLab family [26–28], BiSeNet [29], and PIDNet [30]. Recently, the adoption of vision transformers (ViT) has led to advancements in per-pixel classification models, such as SETR [31], Segformer [32], and Segmenter [33]. Additionally, recent work on mask-classification-based architectures [34–38] has introduced new approaches to semantic segmentation. However, these methods are computationally expensive and not well-suited for deployment on edge devices, which are critical for edge applications with limited computational resources.

In this paper, we propose a Lightweight Attention Network (LANet) for efficient semantic segmentation of curtain wall module frames, which correspond to CWM installation locations. Our architecture overcomes the limitations of existing methods through a novel single-stream design that integrates the strengths of two-pathway and encoder-decoder architectures while eliminating their drawbacks. The network includes a lightweight backbone with MBConv blocks for efficient feature extraction, an Efficient Transformer module for global context modeling, and Channel Attention Modules (CAM) for feature refinement. This design achieves competitive segmentation accuracy while significantly reducing computational costs compared to existing approaches, making it well-suited for real-time deployment in construction environments. The main contributions of this work are as follows:

- (1) A vision system for assisting the manual installation of curtain wall modules has been developed using semantic segmentation.

This approach enables the accurate detection of installation locations, allowing either crane operators or autonomous robots to precisely position the modules during the installation process.

- (2) A lightweight single-stream segmentation architecture has been proposed, integrating an optimized transformer module for effective global context modeling with linear complexity. A remarkable speedup in semantic segmentation of curtain wall module frames has been achieved on mobile CPUs and GPUs, surpassing prior state-of-the-art models.
- (3) A custom curtain wall dataset has been established specifically for automating curtain wall installation. This dataset has been utilized for training and evaluating the LANet model and others, ensuring its suitability for real-world construction environments.
- (4) The feasibility of lightweight models for automating hazardous construction tasks has been validated, bridging critical gaps in robotic systems for semi-automated curtain wall module installation.

The rest of this paper is organized as follows. Section 2 reviews previous works related to our work. Section 3 describes the proposed method in detail. Section 4 evaluates the performance of the proposed method. Lastly, Section 5 concludes the findings of this paper.

## 2. Background

Semantic segmentation is a fundamental computer vision task that aims to partition an input image into regions corresponding to different semantic categories. The goal is to assign a class label to each individual pixel in the image, effectively grouping pixels that belong to the same semantic category.

Recent advances in semantic segmentation have focused on developing efficient architectures suitable for real-time applications. While various approaches exist for optimizing neural networks, including quantization techniques [39], network pruning strategies [40], and knowledge distillation methods [41], our research concentrates specifically on lightweight model design [42]. This approach aims to create inherently efficient architectures through innovative structural designs rather than post-training optimization. Since our method falls under the final category, we provide a brief review of related work in this area.

### 2.1. Lightweight semantic segmentation

Several lightweight semantic segmentation models have been proposed in the literature. Among them, real-time models primarily follow two dominant paradigms: encoder-decoder architectures (Fig. 1-a), which progressively downsample and upsample features, and two-pathway designs (Fig. 1-b), which process spatial and contextual information in parallel.

#### 2.1.1. Encoder-decoder architecture

This architecture has become a fundamental paradigm in semantic segmentation, pioneered by the seminal work of FCN (Fully Convolutional Networks) [21]. These architectures typically consist of two main components: an encoder that progressively downsamples feature maps to capture high-level semantic information, and a decoder that gradually recovers spatial details to generate dense predictions. The encoder can be a lightweight backbone pre-trained on ImageNet [43] or trained from scratch. Significant contributions to this framework include SegNet [22], U-Net [23], and PSPNet [24], each introducing innovative approaches to feature extraction and reconstruction. Similarly, the DeepLab family [26–28] marked a notable advancement by introducing atrous spatial pyramid pooling, enabling multi-scale context capture. OCRNet [25] further refined this approach by incorporating object-contextual representations.

In the context of hybrid CNN and transformer networks, SegFormer [32] unified Transformers with lightweight multilayer perception (MLP) decoders, demonstrating that simple designs can be

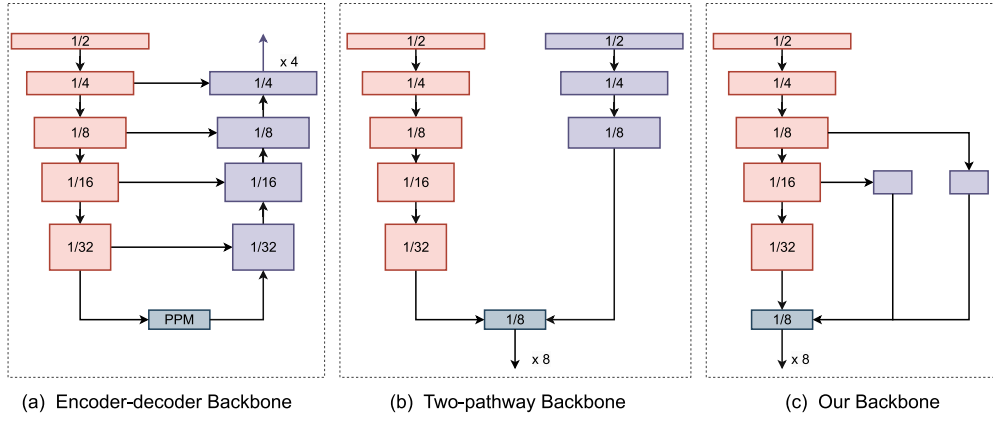


Fig. 1. Comparison of different lightweight semantic segmentation frameworks. (a) The Encoder–Decoder Backbone follows a hierarchical structure, progressively downsampling and upsampling feature maps, with modules like Pyramid Pooling Module (PPM) to enhance multi-scale feature representation. (b) The Two-pathway Backbone processes features through parallel pathways at different resolutions, enabling efficient feature extraction and fusion. (c) Our Backbone integrates hierarchical feature extraction with efficient attention mechanisms.

effective for semantic segmentation. SegNeXt [44] showed that convolutional attention is more efficient than self-attention mechanisms, achieving competitive performance with reduced computational costs. While these architectures effectively balance low-level details and high-level semantics through either dilation backbones or encoder–decoder structures, they often incur substantial computational costs due to high-resolution feature processing and complex network connections.

### 2.1.2. Two-pathway architecture

Two-pathway architectures have emerged as an effective approach for real-time semantic segmentation, offering a balanced trade-off between accuracy and computational efficiency. These architectures typically employ parallel processing paths to capture different aspects of visual information. BiSeNetV1 [45] and BiSeNetV2 [29] proposed a two-pathway architecture by treating spatial details and categorical semantics separately through a detailed branch with wide channels and a lightweight semantic branch. DDRNet [46] proposed a dual-resolution network structure with multiple bilateral fusions, achieving efficient information fusion and maintaining real-time performance with a similar framework. PIDNet [30] proposes a three-pathway framework inspired by proportional–integral–derivative (PID) controllers. In this architecture, each branch is dedicated to a specific function: one branch processes fine-grained details, another extracts deep semantic features, and the third refines boundary information. This innovative extension of the conventional two-pathway paradigm facilitates real-time performance while maintaining high segmentation accuracy.

However, a critical trade-off still exists within the domain of efficient semantic segmentation networks. Multi-branch networks, while capable of extracting comprehensive features, often incur efficiency losses, whereas encoder–decoder structures (e.g., U-Net variants) tend to impose substantial computational costs during the feature reconstruction phase. In contrast, single-stream architectures consolidate these processes into a unified pipeline, thereby reducing model complexity and the total number of parameters. Such simplifications not only improve inference speed but also decrease memory demands, rendering these models particularly well-suited for deployment on edge devices and real-time applications. Our work follows the style of single-stream architectures that offer a balanced trade-off between accuracy and computational efficiency.

### 2.2. Transformer attention mechanisms

Attention mechanisms in computer vision aim to mimic the human visual cognitive system by focusing on essential features within input images [47]. Self-attention, also known as intra-attention, enables input

sequences to interact with each other by encoding relationships between all input entities. This mechanism measures similarities between feature representations by replicating feature maps into three copies (Query, Key, and Value) and computing their relationships. Standard transformers compute attention scores using softmax operations [48]. While transformer-based attention effectively captures long-range dependencies, it faces computational challenges when processing high-resolution images due to quadratic complexity with respect to spatial dimensions. This limitation is particularly significant in dense prediction tasks like semantic segmentation, where maintaining spatial resolution is crucial. Furthermore, while transformers excel at modeling global relationships, they may not effectively preserve local feature details that are important for precise segmentation boundaries.

To address these limitations, various efficient self-attention mechanisms have been proposed in the literature [49–54]. These approaches aim to reduce the computational complexity while maintaining the ability to model long-range dependencies. Linear attention mechanisms, in particular, have shown promise in reducing complexity from quadratic to linear while preserving the effectiveness of self-attention. Our work builds upon these advances by incorporating an efficient linear attention mechanism that enables effective global context modeling while being suitable for deployment in resource-constrained devices.

## 3. Methodology

In this section, we present the detailed architecture of our proposed Lightweight Attention Network (LANet). Building on recent advancements in efficient semantic segmentation, our work extends the principles of the STDC (Short-Term Dense Concatenate) network [55]. Specifically, we leverage its lightweight yet effective feature aggregation strategy to enhance real-time segmentation performance. Our approach introduces critical enhancements that mitigate the limitations inherent to both multi-branch and encoder–decoder architectures discussed in the previous sections. In particular, we augment global context modeling by integrating an Efficient Transformer module at the deepest stage of the network. This integration circumvents the structural redundancy commonly associated with multi-branch frameworks (e.g., PIDNet, BiSeNet, DDRNet), which rely on parallel feature processing, thereby eliminating unnecessary overhead while preserving robust feature representations. Our method consolidates feature extraction and context modeling into a unified framework, as illustrated in Fig. 1-c. By removing parallel branches and redundant operations, our design achieves state-of-the-art accuracy with a minimal parameter count, making it highly suitable for deployment on resource-constrained edge devices.

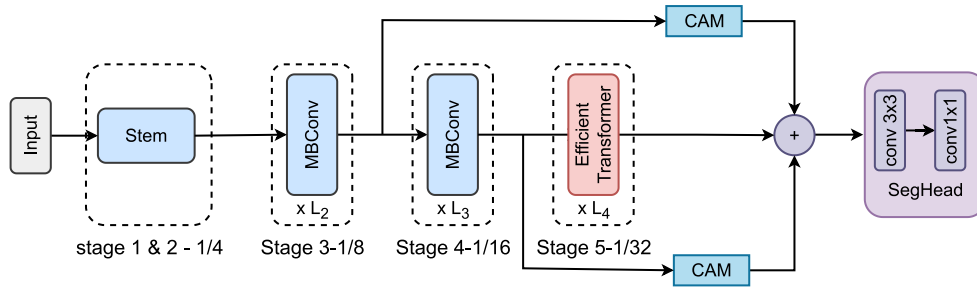


Fig. 2. LANet architecture. The network consists of a stem block for initial feature extraction, followed by MBConv in Stages 3 and 4. We insert an Efficient Transformer module in Stage 5. Channel Attention Modules (CAMs) are integrated to enhance global context awareness, and the SegHead generates pixel-wise predictions.

Table 1

Architectural Details of LANet. The table outlines the configuration of each stage in the network, including the operator type, output size, number of channels, and repetition count. 'C' represents the number of channels, 'H' and 'W' denote the height and width of the feature map, respectively.

| Layers    | Operator              | Output size      | Output channels | Repeat |
|-----------|-----------------------|------------------|-----------------|--------|
| Image     |                       | $384 \times 384$ | 3               |        |
| Stage 1–2 | Stem Block            | $96 \times 96$   | 32              | 1      |
| Stage 3   | MBConv                | $48 \times 48$   | 64              | 4      |
| Stage 4   | MBConv                | $24 \times 24$   | 128             | 3      |
| Stage 5   | Efficient Transformer | $12 \times 12$   | 256             | 1      |

### 3.1. Network architecture

The detailed configuration of our proposed architecture is summarized in Table 1, which outlines the network structure, including operator types, feature map dimensions (height and width), output channels, and the number of repeated blocks at each stage. As illustrated in Fig. 2, LANet comprises three key components: a Lightweight Backbone, an Efficient Transformer module, and a Segmentation Head. The Lightweight Backbone begins with a Stem Block in Fig. 3-a, which is designed to efficiently extract low-level features, followed by MBConv blocks arranged in a progressive downsampling fashion. The backbone consists of five stages, with the final stage incorporating the Efficient Transformer module to capture global dependencies. To enhance feature representation without introducing redundant pathways, Channel Attention Modules (CAMs) are applied to refine features from stages 3 and 4. The Segmentation Head consists of two convolutional layers for final predictions.

The single-stream design eliminates the structural redundancy of multi-pathway architectures with significantly reduced computational cost. As a result, LANet achieves competitive segmentation performance with fewer parameters and computational resources than existing approaches. The entire architecture is optimized for deployment on edge devices. This makes it well-suited for automating curtain wall module installation in dynamic construction environments, where real-time performance and resource constraints are critical.

### 3.2. Channel attention module

We propose a Channel Attention Module (CAM) as shown in Fig. 3-b to enhance feature learning at stage 3 and stage 4 by capturing both global context and pixel-level attention. The CAM first employs global average pooling to extract global contextual information, followed by a pixel-wise attention mechanism to refine features. This can be mathematically formulated as:

$$F_{attn} = \sigma(f_{1 \times 1}(F_{gap})) \quad (1)$$

where  $\sigma$  denotes the sigmoid function,  $F_{gap}$  denotes global pooling and  $f_{1 \times 1}(\cdot)$  represents a  $1 \times 1$  convolutional layer. Finally, the refined features are obtained by:

$$y = F_{attn} * X \quad (2)$$

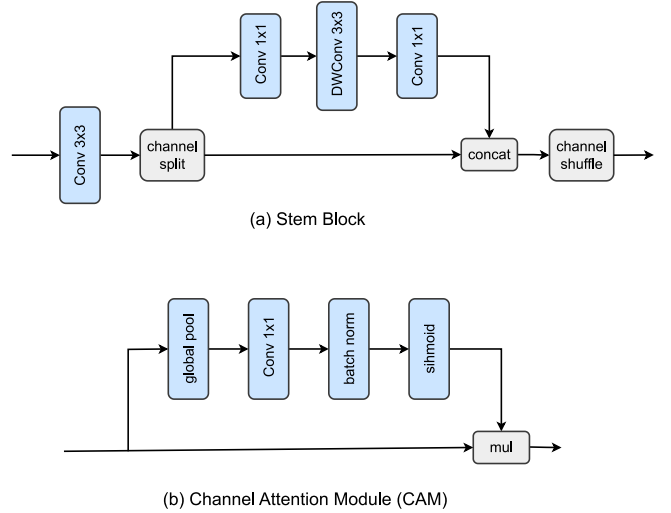


Fig. 3. The complete structure of (a) Stem Block and (b) Channel Attention Module (CAM).

The CAM enhances the network's ability to capture both global semantic information and local detailed features.

### 3.3. Efficient transformer

While Convolutional Neural Networks (CNNs) excel at local feature extraction, they inherently struggle to capture global contextual information crucial for high-level vision tasks. To address this limitation, we incorporate a transformer-based architecture into our design. However, processing high-resolution images directly with traditional transformer modules presents significant computational challenges compared to natural language processing tasks. Therefore, we adopt an Efficient Transformer (ET) module, as shown in Fig. 4, which enables efficient modeling of long-range dependencies while maintaining computational feasibility for dense prediction tasks. Inspired by EfficientViT [56], we substitute the softmax attention with lightweight Multi-Scale Linear Attention (MLA). This module provides an effective means for learning global receptive fields and fusing multi-scale features. It preserves low computational complexity in comparison to the softmax attention mechanism. The softmax attention [48] is given by:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{D}}\right)V \quad (3)$$

where  $Q = xW_Q$ ,  $K = xW_K$ ,  $V = xW_V$ ;  $x \in \mathbb{R}^{n \times f}$  is the input, and  $W_Q, W_K, W_V \in \mathbb{R}^{f \times d}$  are the learnable linear projection matrices.

The attention output for the  $i$ th query in Eq. (3) can be expressed as:

$$O_i = \sum_{j=1}^N \frac{\text{Sim}(Q_i, K_j)}{\sum_{l=1}^N \text{Sim}(Q_i, K_l)} V_j \quad (4)$$

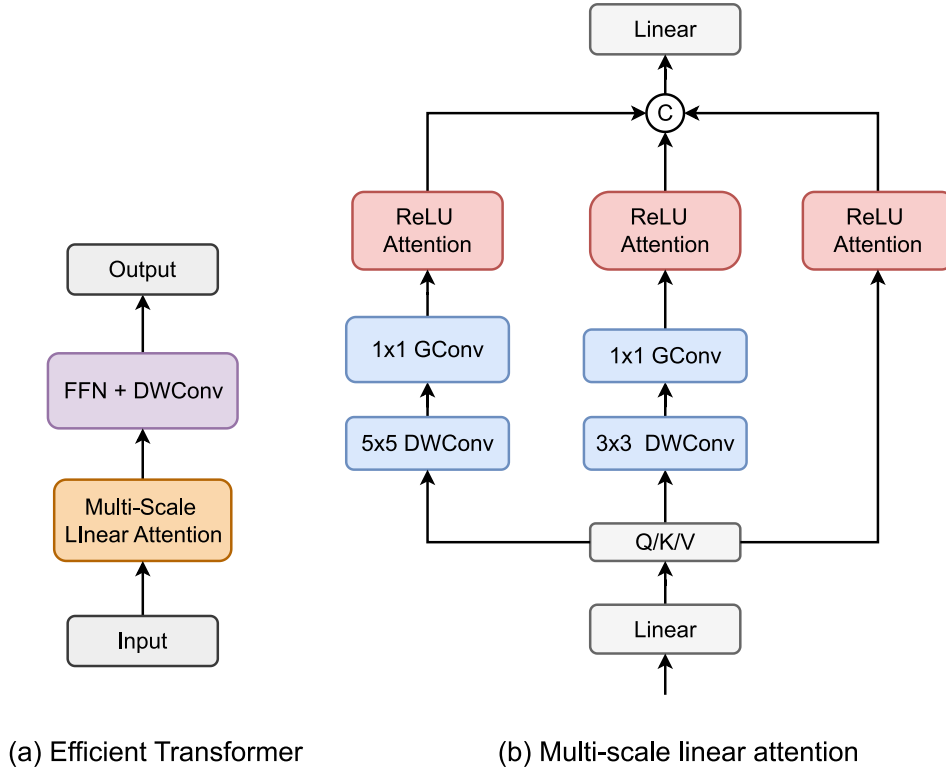


Fig. 4. The Efficient Transformer module in LANet. (a) The overall efficient transformer structure: it consists of multi-scale linear attention followed by FFN and DWConv. (b) Multi-Scale Linear Attention: it employs parallel branches with different kernel sizes ( $5 \times 5$ ,  $3 \times 3$ ) in DWConv layers and  $1 \times 1$  GConv, followed by ReLU attention operations.

When using the similarity function  $Sim(Q, K) = \exp\left(\frac{QK^T}{\sqrt{d}}\right)$ , Eq. (4) becomes the original softmax attention in Eq. (3). If we replace  $Sim(Q, K)$  by  $\phi(Q)\phi(K)^T$  Eq. (4) can be rewritten as:

$$O_i = \sum_{j=1}^N \frac{\phi(Q_i)\phi(K_j)^T}{\sum_{i=1}^N \phi(Q_i)\phi(K_i)^T} V_j \quad (5)$$

Using the associative property, we can simplify further:

$$O_i = \frac{\sum_{j=1}^N [\phi(Q_i)\phi(K_j)^T] V_j}{\sum_{j=1}^N \phi(Q_i)\phi(K_j)^T} \quad (6)$$

Since  $\phi(Q_i)$  is a common factor, we can factorize as:

$$O_i = \frac{\phi(Q_i)(\sum_{j=1}^N (\phi(K_j)^T V_j))}{\sum_{j=1}^N \phi(K_j)^T} \quad (7)$$

This simplification reduces computational complexity by grouping matrix multiplications. Thus, the final equation becomes:

$$O_i = \frac{\phi(Q_i)(\sum_{j=1}^N (\phi(K_j)^T V_j))}{\sum_{j=1}^N \phi(K_j)^T} \quad (8)$$

where  $\phi$  represents the ReLU function, resulting in ReLU linear attention [50]. The ReLU linear attention mechanism reduces the computational complexity from  $O(n^2)$  to  $O(n)$ , where  $n$  is the number of pixels in the feature map, enhancing computational efficiency.

The Efficient Transformer operates on the last feature map of our network through a series of steps. Initially, the input feature map to the transformer is projected into query (Q), key (K), and value (V) representations using linear transformations. We then employ an MLA attention mechanism to avoid the computational complexity of softmax operations. The linear attention operation is applied at multiple scales by using different downsampling ratios on the key and value projections, as shown in Fig. 4-b, allowing the module to capture both local and global contextual information. The integration of the Efficient

Transformer at the final feature map provides several advantages: First, operating on the last feature map enables effective integration of global contextual information. Second, the multi-scale processing within the MLA module facilitates multi-scale feature extraction without introducing additional convolutional layers. Third, by applying the Efficient Transformer at this stage, computational resources are concentrated on refining the most semantically rich features, enhancing both accuracy and efficiency.

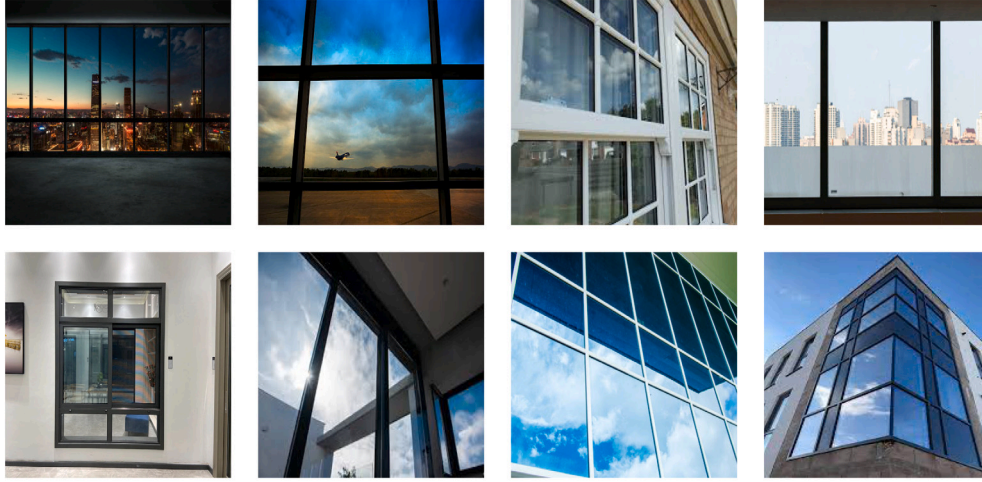
This design allows LANet to strike an optimal balance between model complexity and feature representation capability, making it well-suited for resource-constrained environments. Additionally, the MLA module is designed to be hardware-efficient, avoiding operations that are suboptimal for modern hardware accelerators. This ensures that the theoretical efficiency gains translate into practical speedups across diverse computing platforms, from mobile CPUs to cloud GPUs.

## 4. Experimental results

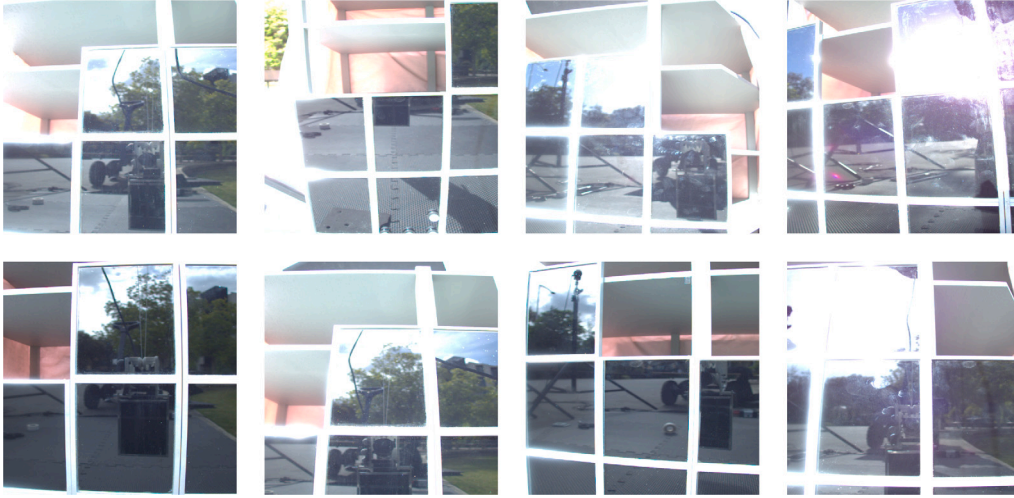
### 4.1. Datasets

The dataset for this study was compiled from two primary sources: publicly available images from open-access repositories and images captured from a lab prototype.

**Multi-class curtain wall dataset.** Our multi-class dataset contains 571 images of a diverse range of curtain wall frame architectural styles and structural elements. It includes 456 images for training and 114 images for validation, with a variety of image resolutions. The finished building images (Fig. 5(a)) capture a variety of real-world architectural styles, ensuring diversity in structural elements. These images also feature different environmental conditions, such as varying lighting levels (e.g., daylight, artificial lighting, and shadows) and occlusions caused by reflections, window mullions, and other structural elements. Meanwhile, the lab prototype images [11] (Fig. 5(b)) provide controlled settings for testing and validation, allowing for a detailed



(a) Finished building



(b) Lab prototype [11]

**Fig. 5.** Dataset preparation: (a) Finished buildings, representing real-world architectural styles and structures. (b) Lab prototypes, providing controlled conditions for model validation and testing.

examination of segmentation performance under different lighting conditions and glass transparency levels. Fig. 6 presents visual examples from our dataset, including both multi-class and binary segmentation masks.

**Binary curtain wall dataset.** To enhance the dataset's applicability and create a more comprehensive benchmark, we adapted our original three-class dataset to a binary classification format, aligning it with the curtain wall frame (CWF) dataset described in [14]. Sample images are shown in Fig. 7. We then combined our dataset with the CWF dataset, resulting in a new benchmark that we refer to as the binary curtain wall dataset. This enhanced dataset incorporates images from both finished and under-construction buildings, offering a more diverse representation of real-world scenarios, with a total of 1,817 images partitioned into training and validation sets. The binary curtain wall dataset offers a more challenging and realistic benchmark for evaluating detection algorithms across various stages of construction, offering a diverse representation of real-world scenarios that autonomous robots might encounter during curtain wall installation processes.

**Implementation details.** We conduct experiments using the MM-Segmentation library [58] and PyTorch [59] as the deep learning

framework. We adopt the mean Intersection over Union (mIoU) as our evaluation metric. All experiments are conducted on an NVIDIA RTX-3090 GPU. The latency is measured on the RTX-3090 GPU using a batch size of 1 and an image size of  $384 \times 384$ . For training, we employ common data augmentation techniques including random horizontal flipping, random scaling (from 0.5 to 2), and random cropping with a size of  $384 \times 384$ . The batch size is set to 16. We use the AdamW optimizer with an initial learning rate of 0.0001 and a weight decay of 0.05. A poly-learning rate decay policy is applied with a power of 0.9. All models are trained from scratch for 57K iterations on the multi-class curtain wall dataset and 182K iterations on the binary curtain wall dataset.

#### 4.2. Ablation study

In this section, we evaluate the impact of each proposed module on the performance of our framework using a multi-class curtain wall dataset. The baseline model is shown in Fig. 8. As presented in Table 2, the baseline model achieves a mIoU of 78.68, with a lightweight parameter count of 0.79 million and an inference speed of 332 FPS.

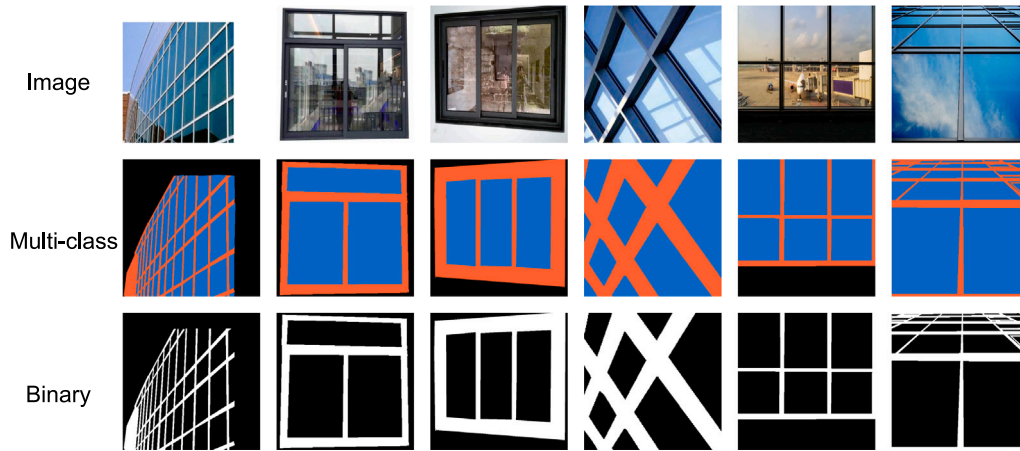


Fig. 6. Samples of the annotated dataset with multi-class and binary segmentation masks.

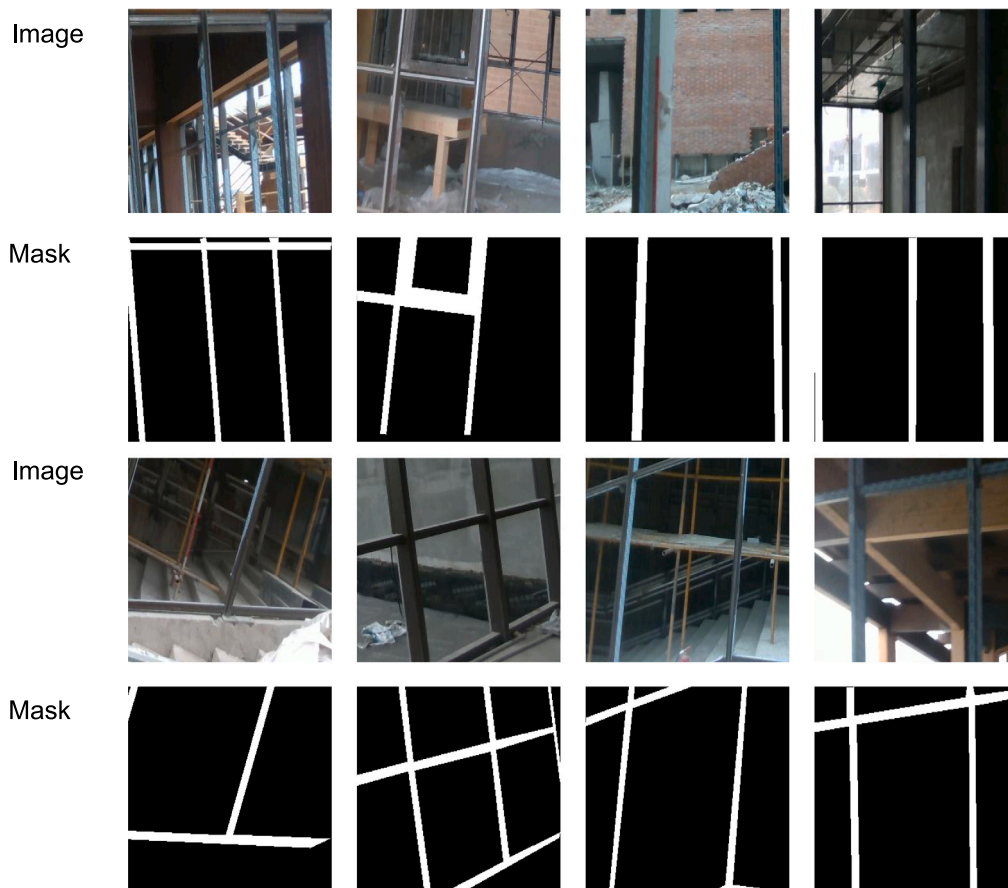


Fig. 7. Examples of curtain wall frames from CWF dataset [57].

**Table 2**

Ablation study evaluating the impact of the proposed modules on the multi-class curtain wall dataset. The table compares the baseline model with the incremental additions of the Efficient Transformer Module and the Channel Attention Module (CAM).

| Method                                 | Parameters(M) | FPS | mIoU  |
|----------------------------------------|---------------|-----|-------|
| Baseline                               | 0.79          | 332 | 78.68 |
| Baseline + Efficient Transformer       | 1.69          | 265 | 83.26 |
| Baseline + Efficient Transformer + CAM | 1.92          | 262 | 84.26 |

**Efficient Transformer:** When the Efficient Transformer module is integrated into the baseline model, the performance improves from an

mIoU of 78.68 to 83.26, demonstrating a more robust capture of the global context. Although the parameter count increases to 1.69 million and the FPS decreases to 265, these changes remain within feasible limits. This trade-off is attributed to the additional attention operations used in extracting better long-range dependencies.

**Channel Attention Module (CAM):** Augmenting the model with CAM alongside the Efficient Transformer leads to a further mIoU improvement from 83.26 to 84.26. CAM adaptively emphasizes the most significant channels, amplifying informative features while suppressing less relevant ones. The final configuration maintains a competitive speed of 262 FPS and 1.92 million parameters. This result highlights the

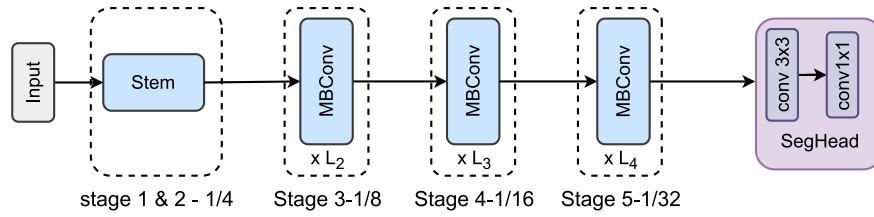


Fig. 8. Architectural overview of the baseline model.

Table 3

Performance comparison of LANet with state-of-the-art lightweight semantic segmentation methods on the multi-class curtain wall dataset. The table evaluates each method across multiple metrics, including per-class Intersection over Union (IoU) for Frame, Glass, and Background categories, mean IoU (mIoU), inference speed in frames per second (FPS), and model size in terms of parameters.

| Method          | Encoder     | Frame | Glass | Background | mIoU ↑       | FPS ↑      | Params ↓    |
|-----------------|-------------|-------|-------|------------|--------------|------------|-------------|
| DDRNet-23 [46]  | –           | 76.45 | 94.63 | 82.92      | 84.66        | 188        | 20.30       |
| BiSeNetV2 [29]  | –           | 76.39 | 94.77 | 83.62      | 84.92        | 201        | 14.77       |
| BiSeNetV1 [45]  | ResNet-18   | 76.76 | 94.18 | 81.03      | 83.99        | 289        | 13.42       |
| DeepLabV3+ [28] | ResNet-18   | 77.05 | 94.64 | 81.93      | 84.54        | 223        | 12.47       |
| HRNet [60]      | HRNetV2-W18 | 77.27 | 94.25 | 79.95      | 83.82        | 34         | 9.64        |
| STDC [55]       | STDC1       | 74.58 | 94.25 | 82.82      | 83.88        | 236        | 8.57        |
| PIDNet-S [30]   | –           | 75.15 | 93.29 | 75.55      | 81.33        | 141        | 7.72        |
| SegNeXt-T [44]  | MSCAN-T     | 76.15 | 94.43 | 83.76      | 84.71        | 93         | 4.23        |
| SegFormer [32]  | MiT-B0      | 76.68 | 93.83 | 78.31      | 82.94        | 149        | 3.72        |
| DeepLabV3+ [28] | MobileNetV2 | 76.34 | 94.63 | 83.87      | 84.95        | 176        | 2.87        |
| LANet (Ours)    | –           | 75.98 | 94.16 | 82.64      | <b>84.26</b> | <b>262</b> | <b>1.92</b> |

Table 4

Performance comparison of LANet with state-of-the-art lightweight semantic segmentation methods on the binary curtain wall dataset. The table evaluates each model based on the mean Intersection over Union (mIoU) and inference speed in frames per second (FPS).

| Method          | Backbone    | mIoU ↑       | FPS ↑      |
|-----------------|-------------|--------------|------------|
| HRNet [60]      | HRNetV2-W18 | 83.32        | 34         |
| SegNeXt [44]    | MSCAN-T     | 82.26        | 93         |
| PIDNet-S [30]   | –           | 82.57        | 141        |
| SegFormer [32]  | MiT-B0      | 82.98        | 149        |
| DeepLabV3+ [28] | MobileNetV2 | 82.57        | 176        |
| DDRNet-23 [46]  | –           | 83.13        | 188        |
| BiSeNetV2 [29]  | –           | 82.29        | 201        |
| DeepLabV3+ [28] | ResNet-18   | 83.30        | 223        |
| STDC [55]       | STDC1       | 81.40        | 236        |
| BiSeNetV1 [45]  | ResNet-18   | 82.47        | 289        |
| LANet (Ours)    | –           | <b>82.85</b> | <b>262</b> |

Table 5

Inference performance comparison of LANet with state-of-the-art lightweight semantic segmentation models on different devices. The table reports the inference speed in frames per second (FPS) on a CPU (13th Gen Intel® Core™ i7-13800H × 20) and a GPU (NVIDIA RTX-3090).

| Method          | Backbone    | CPU       | GPU        |
|-----------------|-------------|-----------|------------|
| DeepLabV3+ [28] | ResNet-18   | 4         | 228        |
| HRNet [60]      | HRNetV2-W18 | 4         | 36         |
| SegNeXt [44]    | MSCAN-T     | 6         | 96         |
| SegFormer [32]  | MiT-B0      | 9         | 151        |
| DeepLabV3+ [28] | MobileNetV2 | 11        | 184        |
| BiSeNetV2 [29]  | –           | 12        | 209        |
| DDRNet-23 [46]  | –           | 13        | 191        |
| BiSeNetV1 [45]  | ResNet-18   | 14        | 287        |
| STDC [55]       | STDC1       | 18        | 235        |
| PIDNet-S [30]   | –           | 19        | 144        |
| LANet (Ours)    | –           | <b>19</b> | <b>262</b> |

complementary nature of the Efficient Transformer and the channel-wise attention strategy in boosting segmentation performance without incurring substantial computational overhead.

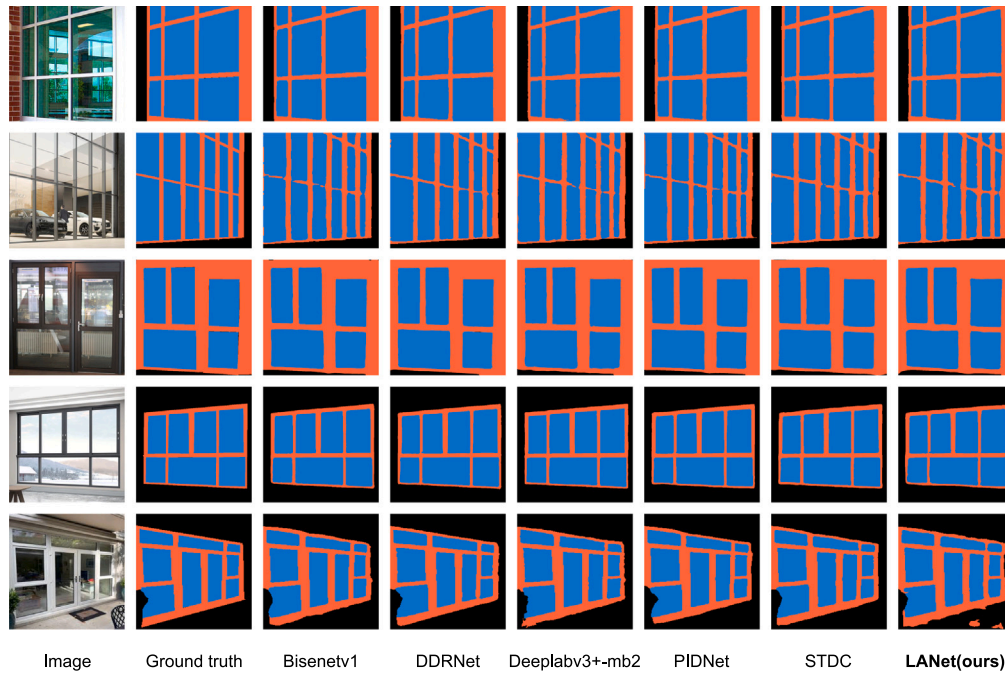
#### 4.3. Comparison with state-of-the-arts

We evaluate our proposed LANet model on both the multi-class and binary curtain wall dataset. The experimental results demonstrate the effectiveness of our approach in terms of accuracy, speed, and computational efficiency.

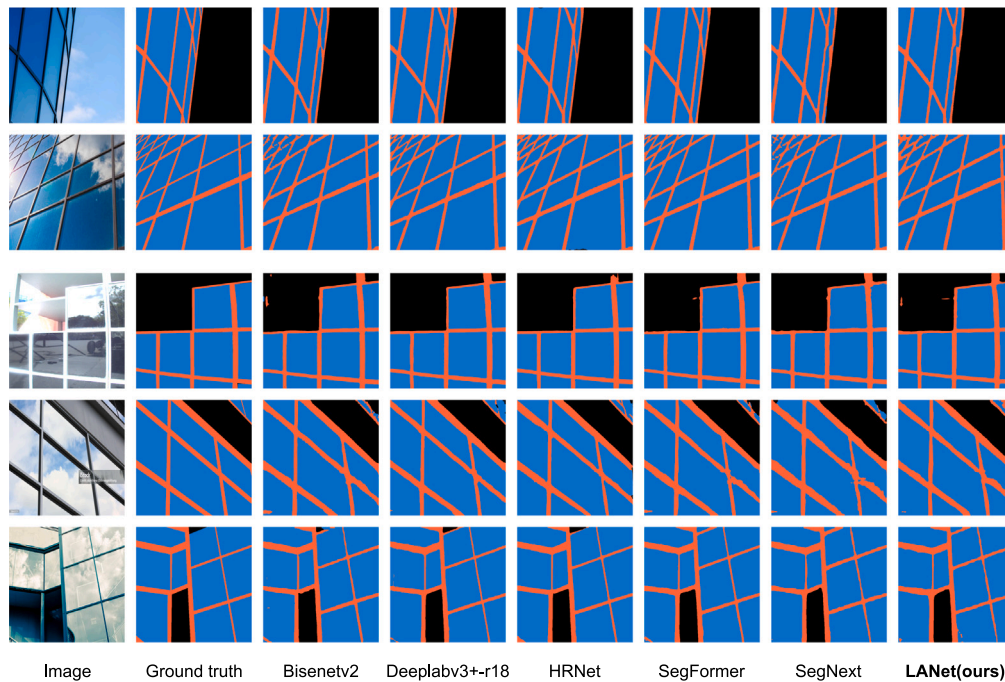
**Evaluation on multi-class curtain wall dataset.** The results on the multi-class curtain wall dataset are presented in Table 3. LANet achieves competitive performance while maintaining significant advantages in computational efficiency. Our model achieves an mIoU of 84.26% on the multi-class segmentation task, on par with state-of-the-art methods such as BiSeNetV2 (84.92%) and DeepLabV3+ (84.95%). However, LANet achieves this performance with only 1.92M parameters, significantly fewer than other models, demonstrating its efficiency. For comparison, BiSeNetV2 requires 14.77M parameters, and DeepLabV3+ with a MobileNetV2 backbone requires 2.81M parameters. Furthermore, LANet operates at 262 FPS, making it one of the fastest models in the comparison, second only to BiSeNetV1 (287 FPS).

**Evaluation on binary curtain wall dataset.** The experimental results for the binary segmentation task on the binary curtain wall dataset are presented in Table 4. LANet achieves an mIoU of 82.85% and maintains a high inference speed of 262 FPS. This performance is comparable to other lightweight models such as BiSeNetV2 (82.29%) and SegFormer-B0 (82.98%), but with a significantly higher inference speed, demonstrating its computational efficiency.

**Inference on different devices.** Table 5 presents the inference performance of various methods on a CPU (13th Gen Intel® Core™ i7-13800H×20) and a GPU (NVIDIA RTX-3090). Among the compared approaches, BiSeNet, DDRNet, DeepLab, FCN, PIDNet, and STDC employ fully convolutional architectures, whereas SegFormer, SegNeXt, and the proposed LANet use hybrid CNN-transformer designs. Despite the typical computational overhead of attention-based frameworks, LANet achieves 19 FPS on a CPU – outperforming all baselines – and 262 FPS on a GPU. These results demonstrate LANet's efficiency in parallel processing on GPUs while maintaining low-latency performance on CPUs. This makes it particularly suitable for edge-device applications, where computational resources and power availability are limited.



**Fig. 9.** Qualitative multi-class segmentation results comparing the proposed LANet (ours) with state-of-the-art methods (BiSeNetV1 [45], DDRNet [46], DeepLabV3+-Mb2 [28], PIDNet [30], STDC [55]).



**Fig. 10.** Qualitative multi-class segmentation results comparing the proposed LANet (ours) with other state-of-the-art approaches (BiSeNetV2 [29], DeepLabV3+-R18 [28], HRNet [60], SegFormer [32], SegNext [44]).

#### 4.4. Qualitative results

To further evaluate the effectiveness of the proposed model, we present qualitative comparisons against several state-of-the-art semantic segmentation methods. Figs. 9 and 10 depict the results for multi-class segmentation, while Figs. 11 and 12 show the performance in binary segmentation tasks. These visual comparisons offer a comprehensive assessment of the model's ability to accurately delineate object boundaries and segment distinct regions.

#### 5. Conclusion

In this paper, we introduced LANet, a lightweight and efficient semantic segmentation model specifically developed for curtain wall frame detection. The proposed architecture incorporates an innovative attention mechanism that offers quadratic improvements in memory and computational efficiency compared to conventional softmax attention. The robust performance of LANet stems from several carefully designed architectural innovations. The lightweight backbone,

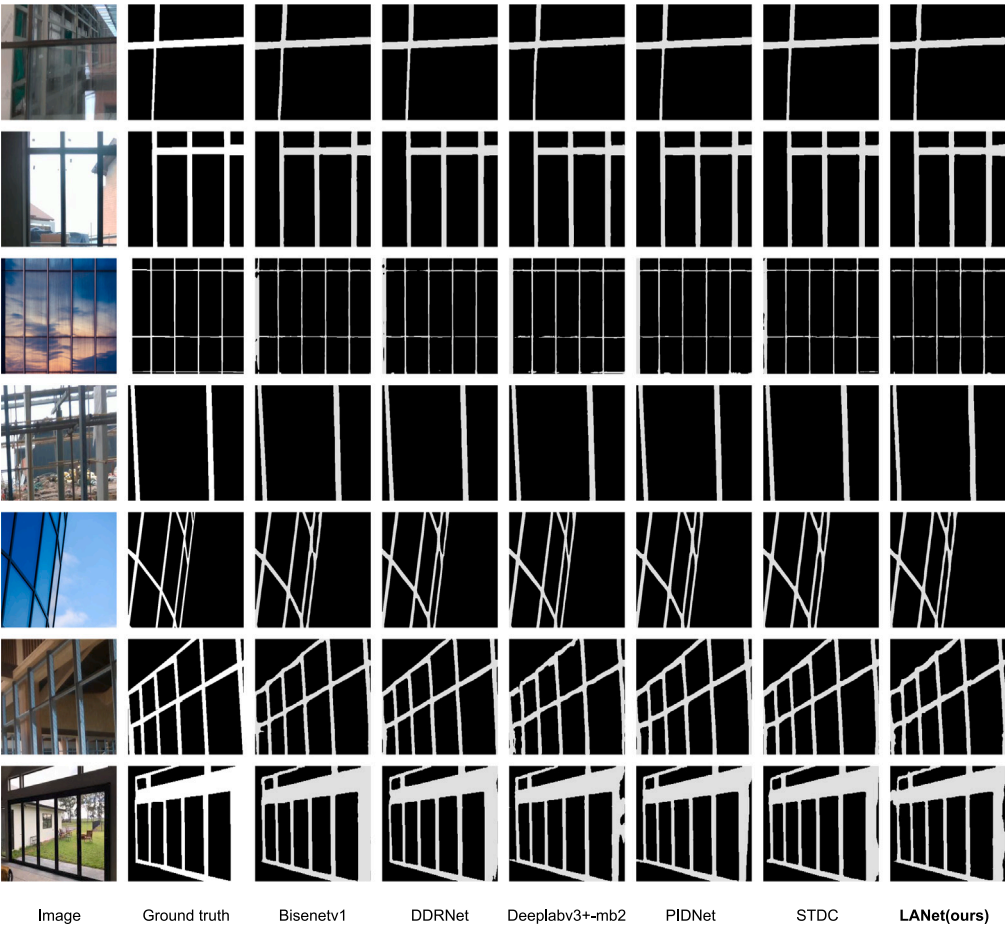


Fig. 11. Qualitative binary segmentation results comparing the proposed LANet (ours) with state-of-the-art methods (BiSeNetV1 [45], DDRNet [46], DeepLabV3+-Mb2 [28], PIDNet [30], STDC [55]).

built with MBConv blocks, effectively reduces the parameter count while preserving strong feature extraction capabilities. In addition, the single-stream architecture eliminates the need for redundant feature extraction pathways, thereby minimizing computational complexity and enhancing inference speed. Experimental evaluations conducted on our custom dataset demonstrate that LANet achieves state-of-the-art efficiency, attaining inference speeds of 19 frames per second on CPUs and 262 frames per second on GPUs. These results affirm the model’s suitability for real-time applications, particularly in edge-device scenarios where computational resources and power budgets are severely constrained.

Future work can explore the sequencing of the positioning, adjusting, and fixing steps in the automated facade installation in high-rise buildings using autonomous robots, contributing to the broader context of robotic assembly and construction automation.

CRediT authorship contribution statement

**Ahmed Yimam Hassen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal

analysis, Data curation, Conceptualization. **Mehrdad Arashpour:** Writing – review & editing, Visualization, Validation, Supervision. **Elahe Abdi:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

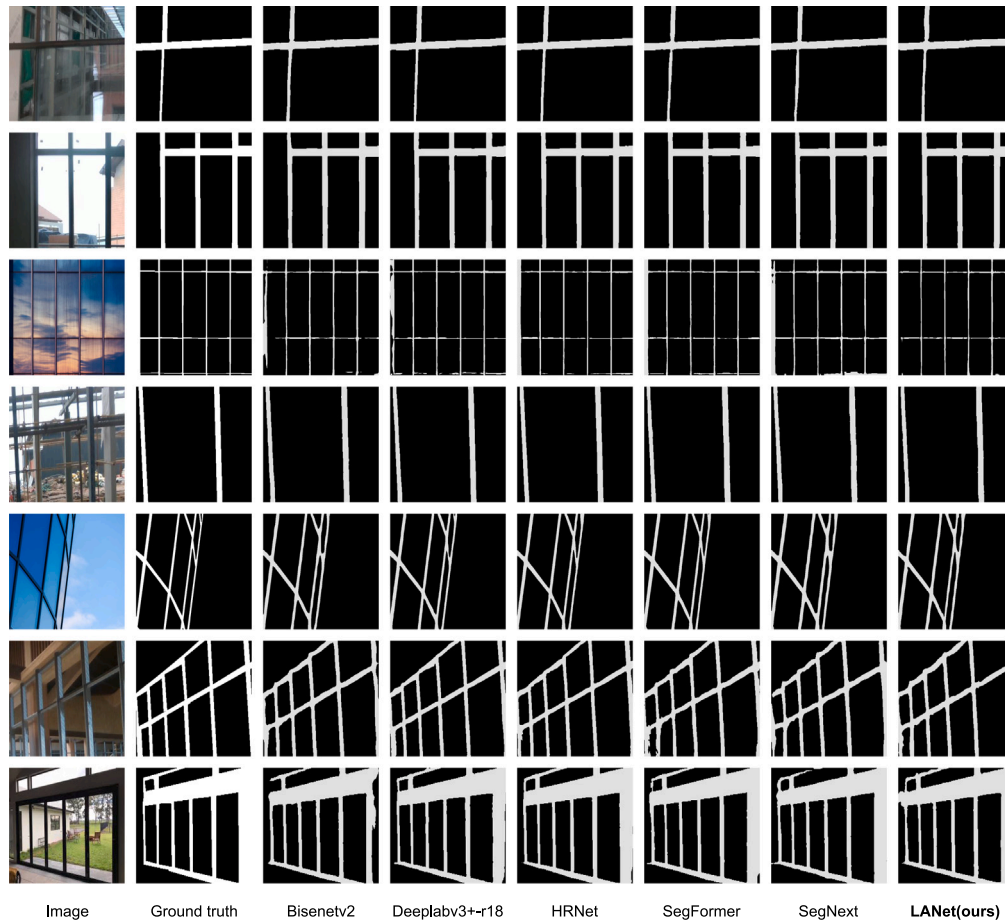


Fig. 12. Qualitative binary segmentation results comparing the proposed LANet (ours) with other state-of-the-art approaches (BiSeNetV2 [29], DeepLabV3+R18 [28], HRNet [60], SegFormer [32], SegNext [44]).

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